



Daisy: the iPhone recycler

Daisy is an Apple-made robot that recycles iPhones, it is part of Apple's initiative to create a healthier planet. <https://digit.in/hidaisy>



Jets race the moon's shadow

During the Jan 2018 lunar eclipse, three F-15 Strike Eagle fighters raced the moon's shadow across the Earth. <https://digit.in/j15jets>

Of ice and fire

Exploring marine clathrates, the co-existence of flammable gas trapped in a cage like structure made of ice

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o quote Captain Nemo, Jules Verne's hero from Twenty Thousand Leagues Under The Sea, "The

sea is a vast reservoir of nature. It was through the sea that the earth, so to speak, began."

Over the last few centuries, the human civilization has unearthed much wealth from the seas and oceans. Would you be surprised if we told you that you can find ice that can catch fire under the oceans? Well, be prepared to be intrigued as we take you on a journey of learning about marine clathrates found in and under the oceanic floor and how it can be of use to humankind.

MARINE CLATHRATES-AN INTRODUCTION

A chemistry textbook would define a clathrate as a molecule or molecules having a cage like structure trapping another component within its crystal structure. It is derived from the Latin word *clatratus* which translates to bars or lattice structures. These clathrates that occur on the ocean floor and are often seen as deposited on seabed are called marine clathrates. In an almost Stockholm syndrome-ish rationalization, the cage structure is called the

host (even if it's trapping its guest which is the molecule inside the lattice). It is rather intriguing that two components exactly opposite in nature and in properties, co-exist in such harmony. Kidnapping? On second thoughts, a marriage more like.

When the cage like structure is made of water molecules and it has methane gas trapped in it, it's called a methane hydrate. In layman's lan-



Naturally occurring Ice entrapping fire

guage it is called as fire ice. This name comes from the fact that, on being subjected to higher temperatures, the ice melts, water drips and methane being highly flammable, catches fire.

It is to be noted that these clathrates hydrates trap not only methane gas but also, a few other gas molecules that are light in weight. Some commonly seen examples are hydrogen, nitrogen, oxygen, argon, xenon etc.

IMPORTANT FACTORS

So what controls the formation and sustenance of these captors and captives? Like most systems that exist on this planet, pressure, temperature and availability of a feasible environment are key factors in the formation of marine hydrates. Abundant water and methane is obviously needed. The main two factors are temperature and pressure. The right concoction of higher pressure and lower temperature has to be reached for methane that is primarily a gas under ambient conditions of pressure and temperature (atmospheric) to form a hydrate and maintain its equilibrium. The type of rock and the composition of this rock also affects their formation. Salt content or salinity along with presence of gas other than methane affect gas hydrates. These gases, commonly, are carbon dioxide, hydrogen sulphide etc.

Where are these exquisite fuel "cubes" found on our planet? If you look at the world map and draw an outline along the continents (well most places, not all, but you get the gist), that's where they occur deposited within certain types of rocks. These locations have the perfect recipe for their occurrence – abundant methane, water, high pressure from the water column being exerted on the raw material and low temperatures in the sea. Interestingly, they are also found on land. Polar Regions are the one place where you will see their existence. They are found below the permafrost.

THE DISCOVERY OF NATURAL METHANE HYDRATES

In the quest to look for alternative fuel sources, the last ten to fifteen years have seen immense research into marine clathrates as a fuel source since its discovery in Russia back in the sixties.

It all started back in the nineteenth century, when hydrates resided within four walls of a laboratory and at the mercy of their human creators. For the longest time, scientists had no reason to believe that hydrates can occur under natural circumstances. This was until the 1930s. The oil and